Predicting Rate of Penetration (ROP) using drilling parameters.

A Comparative Analysis: Statistical Linear Regression vs. Physics-Based Modelling

Statistical Models for Data Science

Mushahid Raza

11-December-2025

Problem Statement & Objective

The Business Problem:

- **Goal:** To predict the Rate of Penetration (ROP) for an oil well to optimize drilling speed and reduce daily rig cost.

Why this matters:

- **Cost:** Offshore rigs cost >$100k/day. Faster drilling = Direct savings.
- **Optimization:** Accurate predictions allow engineers to select the optimal Weight on Bit (WOB) and Rotation Speed (RPM).

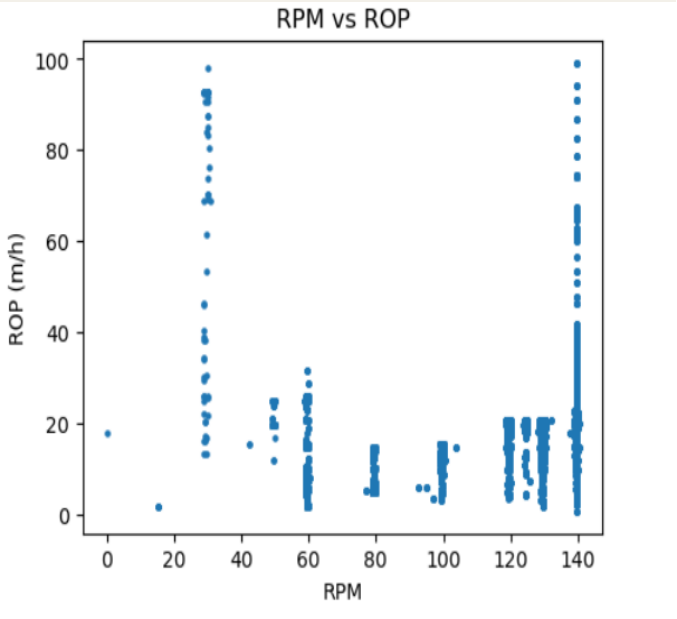
Methodology: We compared two modeling approaches to see which one is better for drilling.

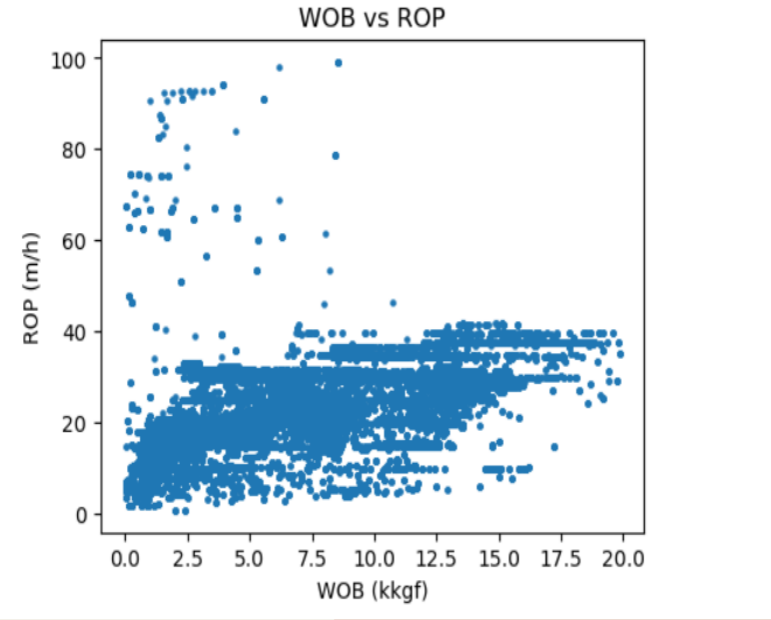
- Statistical Model (Log-Linear Regression): Learns purely from historical data patterns.
- Physics-Based Model (Bingham Model): Uses engineering laws to constrain the predictions.

Dataset Overview & Exploratory Data Analysis

- Source: USROP Public Dataset (GitHub)
- Selected Well: USROP_A3
- Total Observations: ~53,000
- Drilling Variables: ROP, WOB, RPM, SPPA, TFLO, Torque, Depth
- Key drivers: WOB & RPM
- WOB shows strong positive trend with ROP
- RPM shows weak nonlinear relation
- Rest of the variables seemed non-relevant because they are highly corelated.

	Sr No.	MD	WOB	SPPA	Torque	ROP	RPM	TFLO	Mud Weight	Hole Diameter	HKLD	TVD	Gamma gAPI
0	0	1306.525	0.390089	5922.527567	6.521484	70.420992	30.0	1993.927917	11.850475	444.5	121.871198	1283.159232	62.09
1	1	1306.537	0.390089	5922.527567	6.521484	70.420992	30.0	1993.927917	11.850475	444.5	121.871198	1283.159232	62.09
2	2	1306.982	0.839146	6046.633199	7.985768	69.235320	30.0	1993.927917	11.850475	444.5	122.719416	1283.589000	73.36
3	3	1307.135	2.676195	5999.679901	7.443441	91.619832	30.0	1993.927917	11.850475	444.5	120.555780	1283.732256	76.33
4	4	1307.312	6.164320	5960.586628	6.996021	98.069400	30.0	1993.927917	11.850475	444.5	119.771065	1283.875512	62.09





Methodology: The Two Approaches

Model 1 – "Log-Linear Regression" (The Data Approach)

This model looks at historical drilling data (53,000 rows) and finds correlations between the inputs (Weight, Speed) and the output (ROP).

Key Finding:

- Weight on Bit (WOB): Positive coefficient. More weight ≅ Faster drilling.
- RPM: Positive coefficient. Faster spinning ≅ Faster drilling.

Model Accuracy: RMSE: 7.895 m/h.

Why this is a potential problem: The model assumes that if you keep adding weight, the speed will keep increasing forever. It does not account for the physical "saturation point" where the rock cannot break any faster.

Methodology: The Two Approaches

Model 2 – "Bingham Model" (The Physics Approach)

Instead of relying only on patterns in the data, this model includes Engineering Domain Knowledge. It uses a specific formula that accounts for the mechanical efficiency of the drill bit.

Key Finding:

- Coefficient 'a' < 1: The model estimated a parameter of 0.21, which mathematically proves diminishing returns.
- Interpretation: This means that doubling the weight does not double the speed.

Model Accuracy: 7.821 m/h (Better than the Liner Regression Model).

Interpretation: This model aligns with physical reality: using the drill bit too hard eventually stops helping. : p

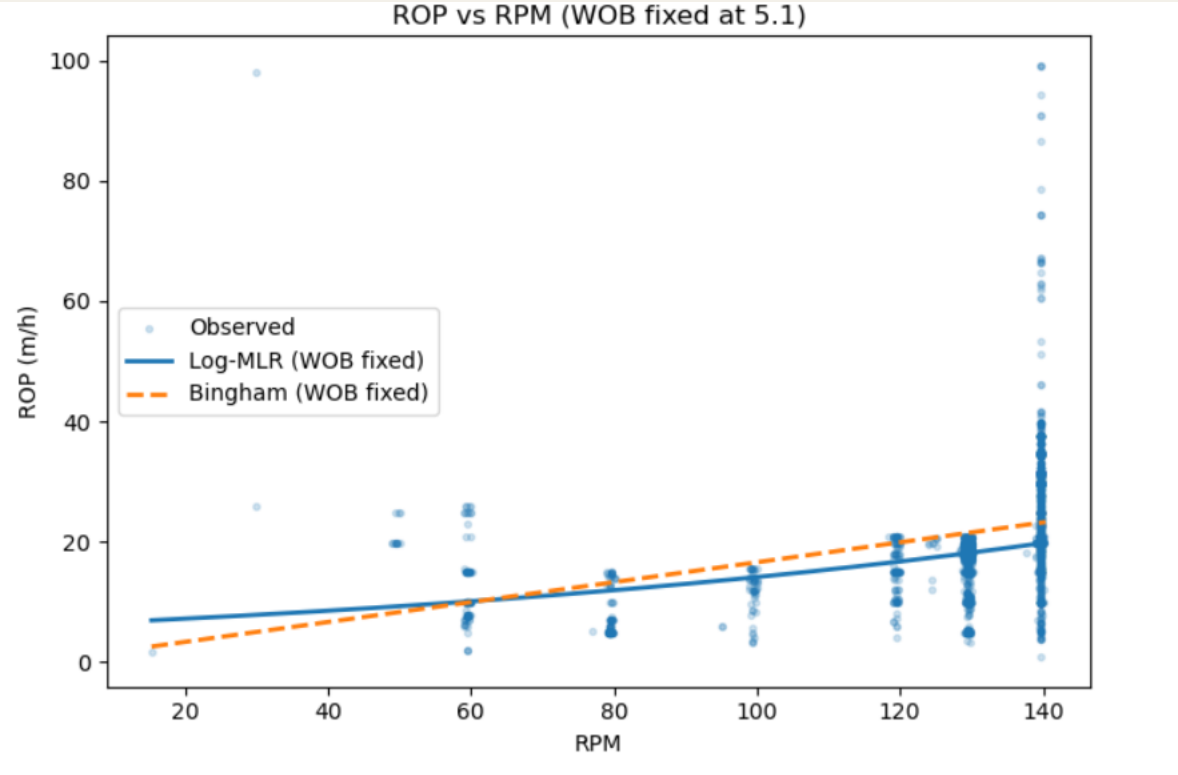
Comparison of Both Models

Similarities in the results: Both models identified Weight on Bit as the main driver of speed. Because the data had a strong signal, both models performed well (RMSE ~7.8).

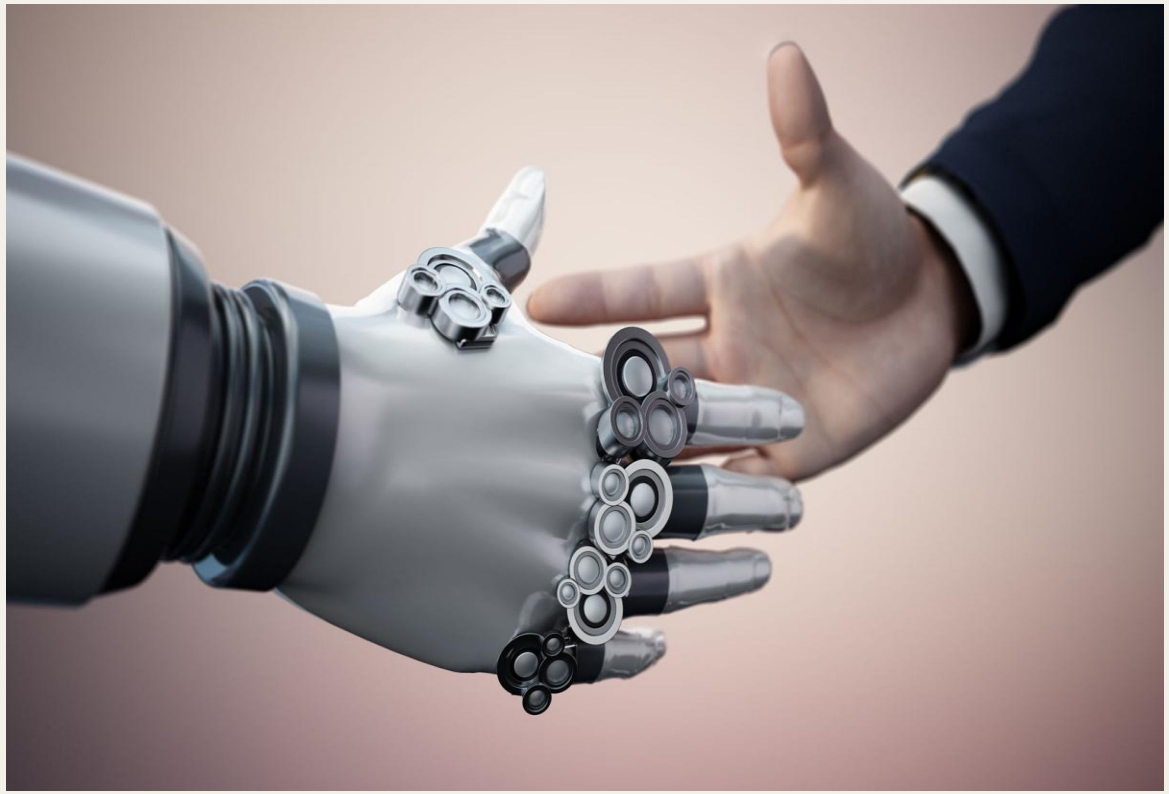
Why the Physics Model is better:

- **The Statistical Model** blindly followed the trend, risking over-prediction at high weights.
- **The Physics Model** included a "structure" (the formula) that acted like a guardrail, preventing unrealistic predictions.

Which model should the engineers rely on?
The **Physics-Based Bingham Model,** because It is not just accurate; it is **safe**. It respects the physical limits of the machinery, whereas the statistical model might recommend dangerous settings just to chase a higher predicted speed.



Conclusion - The Final Takeaway

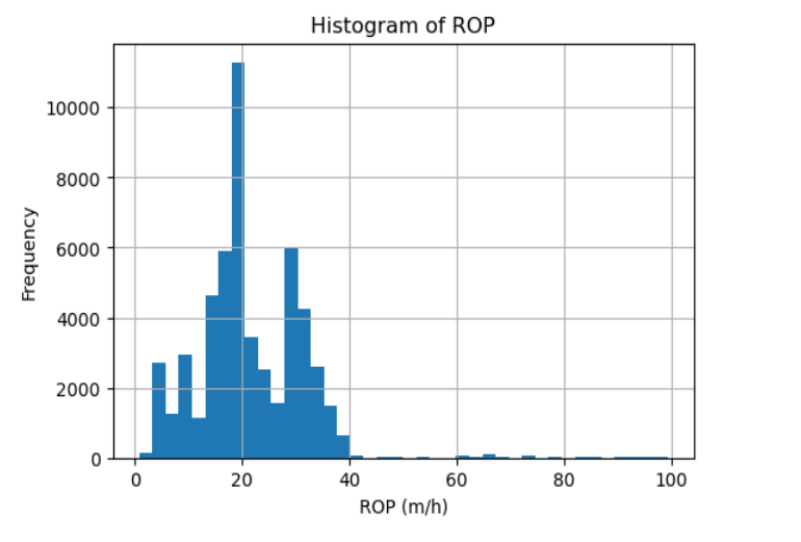


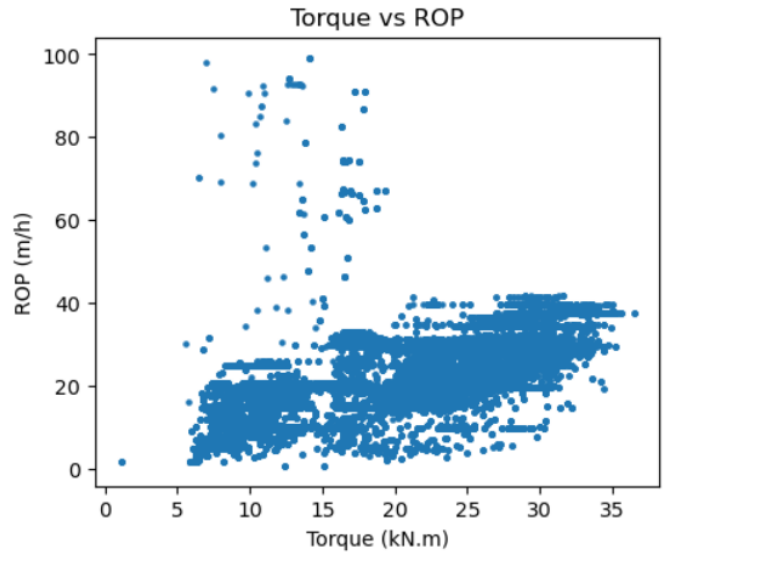
- **Data vs reality:** Data is powerful, but it can be "naive." It doesn't know the physical limits of the machinery.
- **The Winner:** Physics performed better because it understood the real-world constraints.
- **Future Recommendation:** Do not replace the engineering physics. Instead, use data science to tune the physics equations for the best of both worlds.

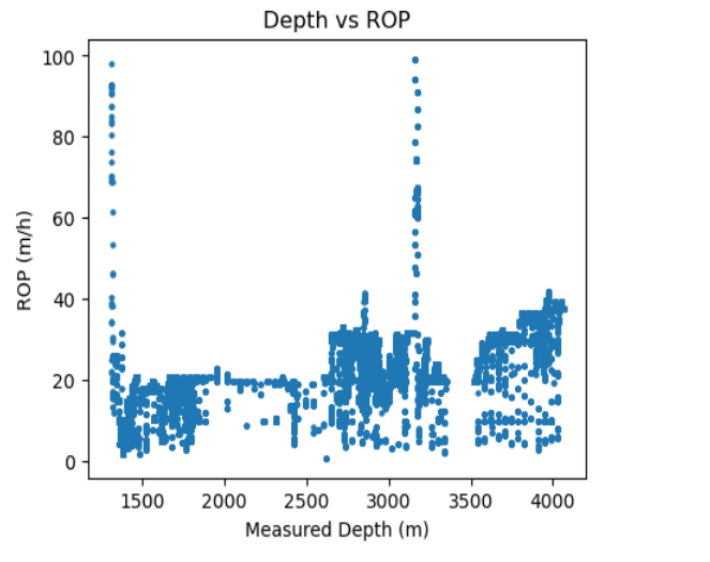
Appendix: EDA

	Sr No.	MD	WOB	SPPA	Torque	ROP	RPM	TFLO	Mud Weight	Hole Diameter	HKLD	TVD	Gamma gAPI
0	0	1306.525	0.390089	5922.527567	6.521484	70.420992	30.0	1993.927917	11.850475	444.5	121.871198	1283.159232	62.09
1	1	1306.537	0.390089	5922.527567	6.521484	70.420992	30.0	1993.927917	11.850475	444.5	121.871198	1283.159232	62.09
2	2	1306.982	0.839146	6046.633199	7.985768	69.235320	30.0	1993.927917	11.850475	444.5	122.719416	1283.589000	73.36
3	3	1307.135	2.676195	5999.679901	7.443441	91.619832	30.0	1993.927917	11.850475	444.5	120.555780	1283.732256	76.33
4	4	1307.312	6.164320	5960.586628	6.996021	98.069400	30.0	1993.927917	11.850475	444.5	119.771065	1283.875512	62.09

	min	max	mean	std
ROP	0.786384	99.206304	21.576512	9.627619
WOB	0.004536	19.858274	6.182876	4.656887
RPM	0.000000	140.351000	130.628762	19.347626
Torque	1.098213	36.489128	19.147319	7.894065
TFLO	1083.309206	4453.120823	2933.600938	1090.260323







•Or residual ACF plots

Appendix: Multicollinearity & Heteroskedasticity

Many drilling variables are **correlated**:

This causes:

- Inflated standard errors & table regression coefficients

We must use :

- Correlation matrix OR Variance Inflation Factor (VIF)

Heteroskedasticity:

- Variance of ROP changes with:
- Depth, Formation changes, and Bit wear

We must run

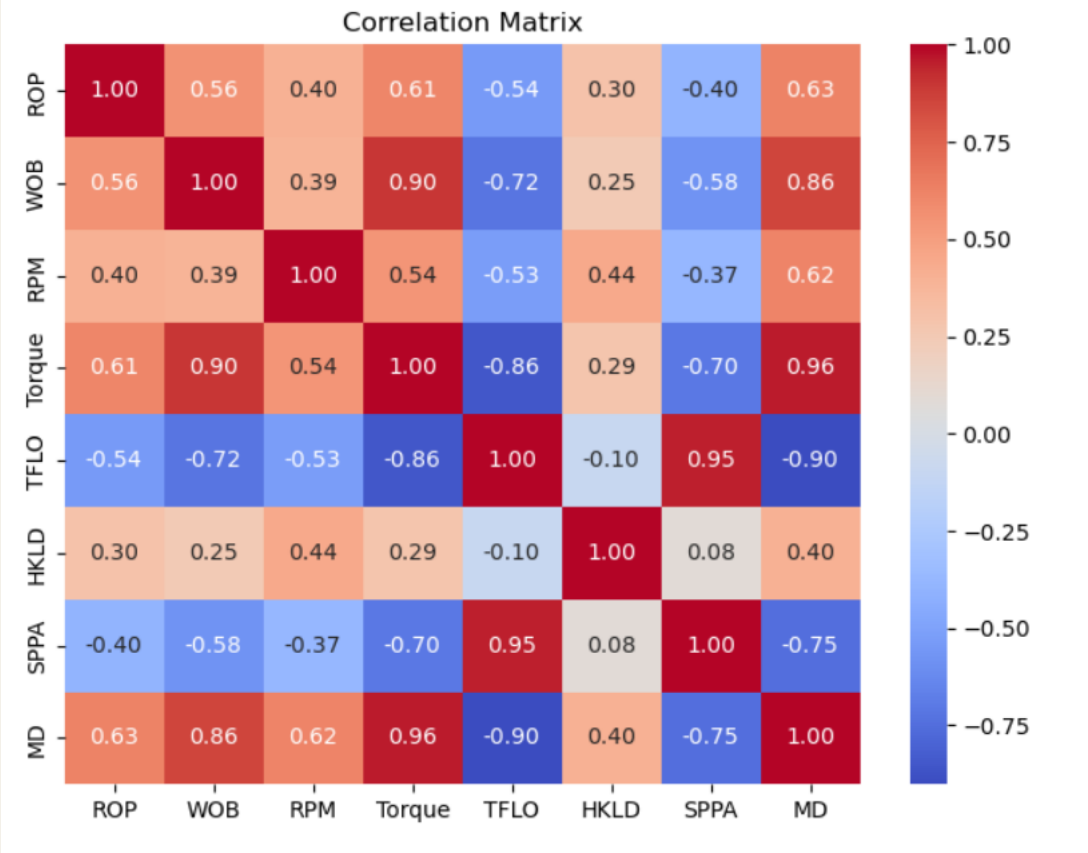
- Breusch–Pagan test or similar Possibly use:
- Robust standard errors OR Weighted least squares

Autocorrelation (Serial Correlation)

- Because data is:
- Depth-ordered & Sequential
- Residuals are not independent:

We have to use

- Check Durbin–Watson Or residual ACF plots



Appendix: Statistical Regression

$$ROP_i = f(X_{i1}, X_{i2}, \ldots, X_{i12}) + \varepsilon_i$$

Where:

•ε_i =random measurement + formation variability

Categories of Features Used :

A) Mechanical Energy / Bit–Rock Interaction Variables

Typical variables in this group:

Weight on Bit (WOB) : How hard the bit presses into the formation. Higher WOB → higher ROP (up to a limit)

Rotational Speed (RPM): How fast the bit rotates. Higher RPM → more cutting action → higher ROP

Torque: Resistance to rotation at the bit. High torque + low ROP → inefficient drilling

Statistical meaning:

These have **direct causal effect** on ROP

They are your **primary regression drivers**

Appendix: Statistical Regression

B) Hydraulic / Circulation Variables:

These control cuttings removal and bottom-hole cleaning.

Typical variables:

- Flow Rate
- Standpipe Pressure
- Mud Density

Proper hydraulics: Clean the bottom hole, Prevent regrinding cuttings, Stabilize ROP

Statistical meaning: These variables often show:

Nonlinear relationships

Interaction effects with WOB and RPM

Appendix: Statistical Regression

C) Well Geometry / Environment Variables

These describe the **drilling environment**, not direct energy.

- Measured Depth / True Vertical Depth
- Inclination
- Hole Size

These affect:

Contact area of the bit

Friction

Cuttings transport efficiency

Statistical meaning: These variables often introduce:

Trend

Heteroskedasticity

Formation-dependent variance

Appendix: Multiple Linear Regression

Initial MLR Model :

$$ROP = \beta_0 + \beta_1 WOB + \beta_2 RPM + \varepsilon$$

- Final Model: (ROP) = 1.17 + 0.97 WOB + 0.11 RPM
- R^2 = 0.352
- ROP in this well is strongly controlled by WOB and moderately controlled by RPM. WOB is roughly 9× stronger than RPM per unit change. Together, they explain about 35% of the total ROP variability, with the remaining variation due to geology, hydraulics, bit wear, and transient conditions.
- F-statistic p-value = 0.00
- The model is globally statistically significant. At least one predictor contributes to explaining ROP

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    ROP   R-squared:                       0.352
Model:                            OLS   Adj. R-squared:                  0.352
Method:                 Least Squares   F-statistic:                 1.438e+04
Date:                Wed, 10 Dec 2025   Prob (F-statistic):               0.00
Time:                        23:04:50   Log-Likelihood:            -1.8389e+05
No. Observations:               53041   AIC:                         3.678e+05
Df Residuals:                   53038   BIC:                         3.678e+05
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          1.1673      0.235      4.974      0.000       0.707       1.627
WOB            0.9726      0.008    123.909      0.000       0.957       0.988
RPM            0.1102      0.002     58.328      0.000       0.106       0.114
==============================================================================
Omnibus:                    33961.240   Durbin-Watson:                   0.026
Prob(Omnibus):                  0.000   Jarque-Bera (JB):          1064704.538
Skew:                           2.601   Prob(JB):                         0.00
Kurtosis:                      24.324   Cond. No.                         922.
==============================================================================
```

Appendix: Log Linear Regression

Initial MLR Model :

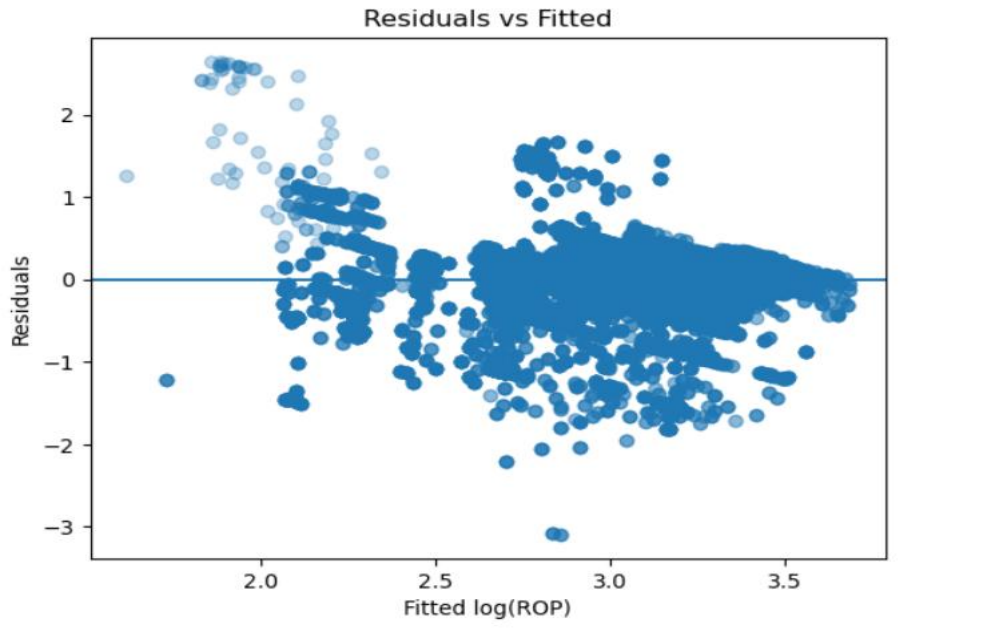
$$ROP = \beta_0 + \beta_1 WOB + \beta_2 RPM + \varepsilon$$

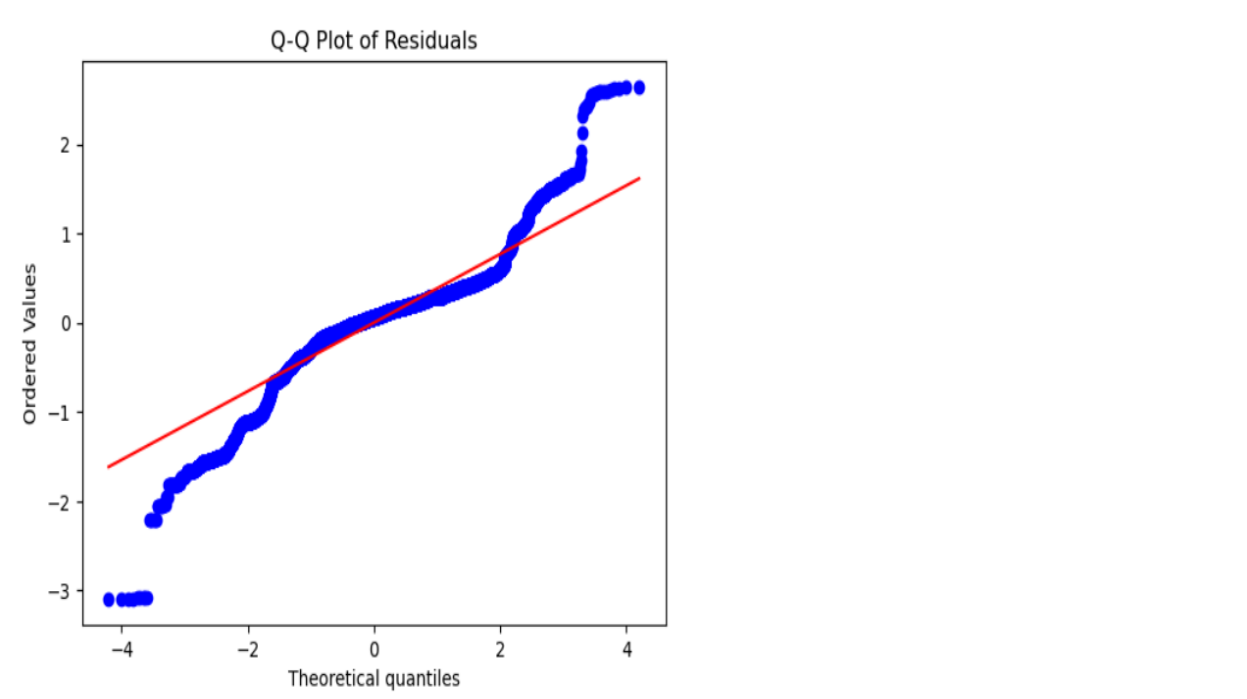
- Final Model: (ROP) = 1.56+ 0.05 WOB + 0.01 RPM
- $R^2 = 0.384$
- A log-linear regression model using WOB and RPM explains 38.4% of the variation in ROP. Both predictors are statistically significant ($p < 0.001$). A 1 kkgf increase in WOB is associated with an average 4.9% increase in ROP, while a 1 rpm increase in RPM increases ROP by approximately 0.84%. The log transformation substantially reduces skewness and heavy-tailed behavior in the residuals, improving the validity of statistical inference. However, strong residual autocorrelation remains due to the time-series nature of drilling data.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                log_ROP   R-squared:                       0.384
Model:                            OLS   Adj. R-squared:                  0.384
Method:                 Least Squares   F-statistic:                 1.654e+04
Date:                Wed, 10 Dec 2025   Prob (F-statistic):               0.00
Time:                        23:05:02   Log-Likelihood:                -27685.
No. Observations:               53041   AIC:                         5.538e+04
Df Residuals:                   53038   BIC:                         5.540e+04
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          1.5612      0.012    126.479      0.000       1.537       1.585
WOB            0.0475      0.000    115.040      0.000       0.047       0.048
RPM            0.0084   9.94e-05     84.899      0.000       0.008       0.009
==============================================================================
Omnibus:                    11138.882   Durbin-Watson:                   0.027
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            56195.421
Skew:                          -0.929   Prob(JB):                         0.00
Kurtosis:                       7.687   Cond. No.                         922.
==============================================================================
```

Appendix: Log Linear Regression

- Model Diagnostics. Residual diagnostics were performed on the log-linear regression model to verify the key OLS assumptions.
- residuals are centered around zero with no strong nonlinear pattern, indicating that the linearity assumption is reasonably satisfied. However, the spread of residuals varies with the fitted values, suggesting heteroskedasticity.
- The Q–Q plot of residuals shows that the log-transformation substantially improves normality compared to the raw ROP model, although deviations remain in the tails, indicating heavy-tailed behavior.
- The presence of heteroskedasticity is formally confirmed by the Breusch–Pagan test (p-value < 0.001), leading to rejection of the null hypothesis of constant variance. In addition, the Durbin–Watson statistic indicates strong positive autocorrelation, which is expected given the time-series nature of drilling data. While these assumption violations limit strict classical inference, the coefficient estimates remain unbiased, and the model is suitable for predictive and operational analysis with the stated limitations.





Breusch-Pagan Test Results:
{'LM Stat': np.float64(1598.966864359625), 'LM p-value': np.float64(0.0), 'F-Stat': np.float64(824.2870604306469), 'F p-value': np.float64(0.0)}

Appendix: Bingham Model – Nonlinear

$$ROP = K\left(\frac{WOB}{D_b}\right)^{a} N$$

Final Bingham Model:

$$ROP = 0.4287\left(\frac{WOB}{444.5}\right)^{0.2126} \cdot RPM$$

Where:

- **ROP** = Rate of Penetration (m/h)
- **WOB** = Weight on Bit
- **Db** = Bit diameter
- **N** = Rotary speed (RPM)
- **K** = constant related to formation strength = 0.43
- **a** = bit weight exponent = 0.21
- a < 1 → diminishing returns of WOB

Appendix: Model Comparison (Using RMSE)

- RMSE (Bingham Model) = 7.821
- RMSE (Log-MLR Model) = 7.895

Numerical Comparison:

$$\Delta RMSE = 7.895 - 7.821 = 0.074$$

That means:

- The Bingham model has a LOWER prediction error.
- The improvement is:
- Small in absolute terms, But consistent and real